

FDN Class

1.0

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Chapter 1

Class Index

1.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

FDNreverb	
Creates a reverb effect using a feedback delay network	5

Chapter 2

File Index

2.1 File List

Here is a list of all documented files with brief descriptions:

C:/JUCE Projects/FDN Class/ FDNreverb.h	9
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Chapter 3

Class Documentation

3.1 FDNreverb Class Reference

Creates a reverb effect using a feedback delay network.

```
#include <FDNreverb.h>
```

Public Member Functions

- void **setroomarea** (double roomarea)
Set the room area parameter.
- void **setabscoef** (double abscoef)
Set absorption coefficient.
- void **setreflection** (double reflection)
Set reflection parameter.
- void **setsamplingrate** (double samplingrate)
Set the sampling rate.
- void **reset** ()
Clear all delay lines.
- double **step** (double sample)
Step FDN reverb through one sample period.
- void **procBlock** (float *samples, unsigned numSamples, double pctReverb=100.0, double dbGain=0.0)
Process a block of audio samples through the FDN reverb.

3.1.1 Detailed Description

Creates a reverb effect using a feedback delay network.

3.1.2 Member Function Documentation

3.1.2.1 procBlock()

```
void FDNreverb::procBlock (
    float * samples,
    unsigned numSamples,
    double pctReverb = 100.0,
    double dbGain = 0.0)
```

Process a block of audio samples through the FDN reverb.

Parameters

<i>samples</i>	pointer to the array of input/output samples
<i>numSamples</i>	number of samples in the array
<i>pctReverb</i>	percent reverb (0-100)
<i>dbGain</i>	gain in dB

3.1.2.2 setabscoef()

```
void FDNreverb::setabscoef (
    double abscoef)
```

Set absorption coefficient.

Parameters

<i>abscoef</i>	HF absorption coefficient in dB/m
----------------	-----------------------------------

3.1.2.3 setreflection()

```
void FDNreverb::setreflection (
    double reflection) [inline]
```

Set reflection parameter.

Parameters

<i>reflect</i>	reflection parameter (no units)
----------------	---------------------------------

3.1.2.4 setroomarea()

```
void FDNreverb::setroomarea (  
    double roomarea)
```

Set the room area parameter.

Parameters

<i>roomarea</i>	room area in m ²
-----------------	-----------------------------

3.1.2.5 setsamplingrate()

```
void FDNreverb::setsamplingrate (  
    double samplingrate) [inline]
```

Set the sampling rate.

Parameters

<i>samplingrate</i>	Sampling rate in Hz
---------------------	---------------------

3.1.2.6 step()

```
double FDNreverb::step (  
    double sample)
```

Step FDN reverb through one sample period.

Parameters

<i>sample</i>	input sample
---------------	--------------

Returns

output sample

The documentation for this class was generated from the following files:

- C:/JUCE Projects/FDN Class/FDNreverb.h
- C:/JUCE Projects/FDN Class/FDNreverb.cpp

Chapter 4

File Documentation

4.1 FDNreverb.h

```
00001 /*
00002  =====
00003
00004     FDNreverb.h
00005     Created: 24 Nov 2019 12:35:34pm
00006     Author:  profw
00007
00008  =====
00009 */
00010
00011 #pragma once
00012
00013 constexpr auto SPEEDOFSOUND = 343.0; // speed of sound in m/s
00014 constexpr auto ORDER = 6;
00015 constexpr double PHI = 1.618; // golden ratio
00016 constexpr auto MAX_SAMPLE_DELAY = 32000; // maximum delay in samples (for delay line buffer size)
00017
00021 class FDNreverb
00022 {
00023 public:
00024     FDNreverb();
00025     ~FDNreverb() {}
00026
00031     void setroomarea(double roomarea);
00032
00037     void setabscoef(double abscoef);
00038
00043     void setreflection(double reflection) { m_reflection = reflection; }
00044
00049     void setsamplingrate(double samplingrate) { m_samplingRate = samplingrate; }
00050
00054     void reset();
00055
00061     double step(double sample);
00062
00070     void procBlock(float* samples, unsigned numSamples, double pctReverb = 100.0, double dbGain =
00071         0.0);
00072 private:
00073     float m_delayLine[ORDER][MAX_SAMPLE_DELAY];
00074     unsigned m_sampleDelay[ORDER];
00075     double m_damping[ORDER];
00076     double x[ORDER];
00077     double m_alpha; // used to determine path lengths
00078     double m_samplingRate;
00079     double m_absCoef; // HF absorption coefficient in dB/m
00080     double m_reflection; // reflection parameter (no units)
00081     unsigned m_oldest; // index of oldest sample in delay line (for circular buffer)
00082 };
```

